

# Statistical Foundations of the Volatility Term Structure Model

COMP 653 Statistical Machine Learning, Summer 2026

## Contents

|          |                                                          |          |
|----------|----------------------------------------------------------|----------|
| <b>1</b> | <b>Problem Definition</b>                                | <b>3</b> |
| 1.1      | The Project Goal . . . . .                               | 3        |
| 1.2      | What Is Being Predicted . . . . .                        | 3        |
| 1.3      | Why Volatility Clusters in Time . . . . .                | 4        |
| 1.4      | The Horizon and the Term Structure . . . . .             | 4        |
| 1.5      | From Volatility to a Range of Gains and Losses . . . . . | 5        |
| <b>2</b> | <b>Solution Design and Key Features</b>                  | <b>6</b> |
| 2.1      | Target Choice That Matches the Signal . . . . .          | 7        |
| 2.2      | Multi Scale Term Structure With Drift . . . . .          | 7        |
| 2.3      | Complementary Models Under One Roof . . . . .            | 7        |
| 2.4      | A Calibrated Range of Prices, Not a Point . . . . .      | 7        |
| 2.5      | Robust Training and Honest Evaluation . . . . .          | 8        |
| <b>3</b> | <b>The Model Lifecycle from Data to Tuning</b>           | <b>9</b> |
| 3.1      | Data . . . . .                                           | 10       |
| 3.2      | Splitting With a Purge and an Embargo . . . . .          | 10       |
| 3.3      | The Training Loop . . . . .                              | 10       |
| 3.4      | Validation and Feedback . . . . .                        | 10       |
| 3.5      | Testing . . . . .                                        | 11       |
| 3.6      | Reporting and Saving . . . . .                           | 11       |
| 3.7      | Tuning and Modifying . . . . .                           | 11       |

|           |                                                            |           |
|-----------|------------------------------------------------------------|-----------|
| <b>4</b>  | <b>Data, Returns, and the Prediction Target</b>            | <b>12</b> |
| 4.1       | Cross Sectional Median Labels . . . . .                    | 12        |
| <b>5</b>  | <b>Feature Standardization</b>                             | <b>14</b> |
| <b>6</b>  | <b>Information Theory Feature Selection</b>                | <b>15</b> |
| 6.1       | Entropy . . . . .                                          | 15        |
| 6.2       | Mutual Information . . . . .                               | 15        |
| <b>7</b>  | <b>Branch A: Logistic Regression</b>                       | <b>18</b> |
| <b>8</b>  | <b>Branch B: Gaussian Naive Bayes</b>                      | <b>20</b> |
| <b>9</b>  | <b>Branch C: The Multilayer Perceptron</b>                 | <b>21</b> |
| <b>10</b> | <b>Branch E: The LSTM and Backpropagation Through Time</b> | <b>22</b> |
| 10.1      | Forward Recurrence . . . . .                               | 22        |
| 10.2      | Backward Recurrence . . . . .                              | 22        |
| <b>11</b> | <b>The Softmax Meta Layer and Cross Entropy</b>            | <b>24</b> |
| <b>12</b> | <b>The Adam Optimizer</b>                                  | <b>26</b> |
| <b>13</b> | <b>Multi Scale Fusion and the Drift Gradient</b>           | <b>28</b> |
| 13.1      | The Curvature Coupling Penalty . . . . .                   | 28        |
| <b>14</b> | <b>Quantile Price Bands</b>                                | <b>30</b> |
| 14.1      | The Pinball Loss and the Quantile . . . . .                | 30        |
| 14.2      | The Monotone Non Crossing Construction . . . . .           | 31        |
| <b>15</b> | <b>Conformal Calibration of the Bands</b>                  | <b>33</b> |
| <b>16</b> | <b>Evaluation by the Area Under the ROC Curve</b>          | <b>36</b> |
| <b>17</b> | <b>The Ensemble</b>                                        | <b>38</b> |
| <b>18</b> | <b>Summary of the Pipeline</b>                             | <b>39</b> |

# 1 Problem Definition

This project predicts the volatility of a stock with respect to time. Volatility is the size of the moves a stock makes, so it governs how far the price can travel up or down over a horizon rather than the sign of that travel. We predict this quantity as a function of the horizon, which is what we mean by with respect to time, and we show that it carries a clear and learnable training signal precisely because volatility clusters intrinsically in time.

## 1.1 The Project Goal

Our goal is stated in one sentence. Given a stock, a target price, and a time horizon, we predict how far the price will move and we report the probability that it reaches the target by a target date, together with the full range of prices the stock can reach at each horizon.

The most ambitious form of this goal is a fully specified claim about a single name. We would like to say that stock  $X$  reaches price  $Y$  by date  $D$  at time  $T$ , then holds above a level  $P$ , and delivers a total gain of  $M$  dollars on a position. We treat that sentence as the target the project aims at, and we build every component toward answering it.

Markets are close to efficient in direction, so we cannot promise a single deterministic path to that claim, and it would be dishonest to pretend otherwise. What we can deliver is the honest probabilistic version of the same statement. For the event that stock  $X$  trades at or above price  $P$  by date  $D$  we estimate

$$\widehat{\Pr}(P_{X,D} \geq P \mid \mathcal{F}_t),$$

and we surround it with a calibrated price cone whose width is set by the predicted volatility, so the target price, the date, and the level all read off one forecast. A wider cone means a larger span of possible gain  $M$ , and a tighter cone means a more confident claim. In this way the ultra specific goal becomes a probability and an interval that we can measure and trust, rather than a certainty that no model can provide.

## 1.2 What Is Being Predicted

We write  $\mathcal{F}_t$  for the information available at time  $t$ , which is every price, return, and feature up to and including day  $t$ . Over the next  $h$  trading days the stock earns a cumulative log return, and the object we predict is the conditional volatility of that return, which is the

standard deviation of the future path given what is known today,

$$\sigma_{t,h} = \sqrt{\text{Var}\left(\sum_{k=1}^h r_{t+k} \mid \mathcal{F}_t\right)}.$$

The reason we target the conditional volatility rather than the raw return is a basic fact about markets. The expected next return given the past is close to zero, which is the efficient market statement that the sign of a move is almost a coin flip,

$$\mathbb{E}[r_{t+1} \mid \mathcal{F}_t] \approx 0.$$

Because the mean is unpredictable the direction carries almost no signal, and in the experiments the direction classifier scores an area under the curve near one half. The magnitude of the move is a different story, and that is where the signal lives.

### 1.3 Why Volatility Clusters in Time

Volatility is predictable because large moves follow large moves and calm follows calm, which is the property called volatility clustering. Stated in statistics, the squared returns are positively autocorrelated and that autocorrelation decays only slowly, so today's turbulence forecasts tomorrow's,

$$\text{Corr}(r_t^2, r_{t-k}^2) > 0 \quad \text{for many lags } k.$$

This persistence is intrinsic to how information and trading arrive in clusters, and it means the conditional variance is a smooth and slowly changing process rather than noise. That is exactly the property a learner can exploit, so the volatility label provides a clear tone for the training signal while the direction label does not. The measured contrast is stark, with the direction target near an area under the curve of one half and the volatility target near one.

### 1.4 The Horizon and the Term Structure

Predicting with respect to time means predicting at many horizons at once. The conditional volatility depends on how far ahead we look, and the collection of these values across horizons is the volatility term structure,

$$h \longmapsto \sigma_{t,h}, \quad h \in \{1, 5, 10, 30, 90, 180\} \text{ days}.$$

Because volatility clusters, longer horizons average over more of the persistent component and become steadier and more predictable, which is why the area under the curve rises with the horizon in the results. The model reads six time windows in parallel and couples their predictions into one smooth curve, so it learns the whole term structure rather than a single horizon in isolation.

## 1.5 From Volatility to a Range of Gains and Losses

Although volatility does not tell us the sign of the next move, it tells us how wide the range of outcomes is, which is what a trader cares about when sizing a position. We turn the predicted volatility into an explicit range of future prices by learning quantiles of the return, so the model reports a cone of possible gains and losses whose width is set by the predicted volatility. A wider cone means a more volatile stock and a larger span of possible gain or loss, which is the practical reading of the forecast.

In short the problem is to forecast, at each horizon, how volatile a stock will be relative to its peers, to exploit the fact that volatility clusters in time, and to express the forecast as a calibrated range of future gains and losses.

## 2 Solution Design and Key Features

Our design solves the problem by attacking it at every scale of time at once, by learning the target with a set of complementary models, and by turning the forecast into a calibrated range rather than a single number. This section walks through the design and maps each choice to the part of the problem it answers, and the sections that follow derive the formulas behind each piece. The whole pipeline is shown in Figure 1.

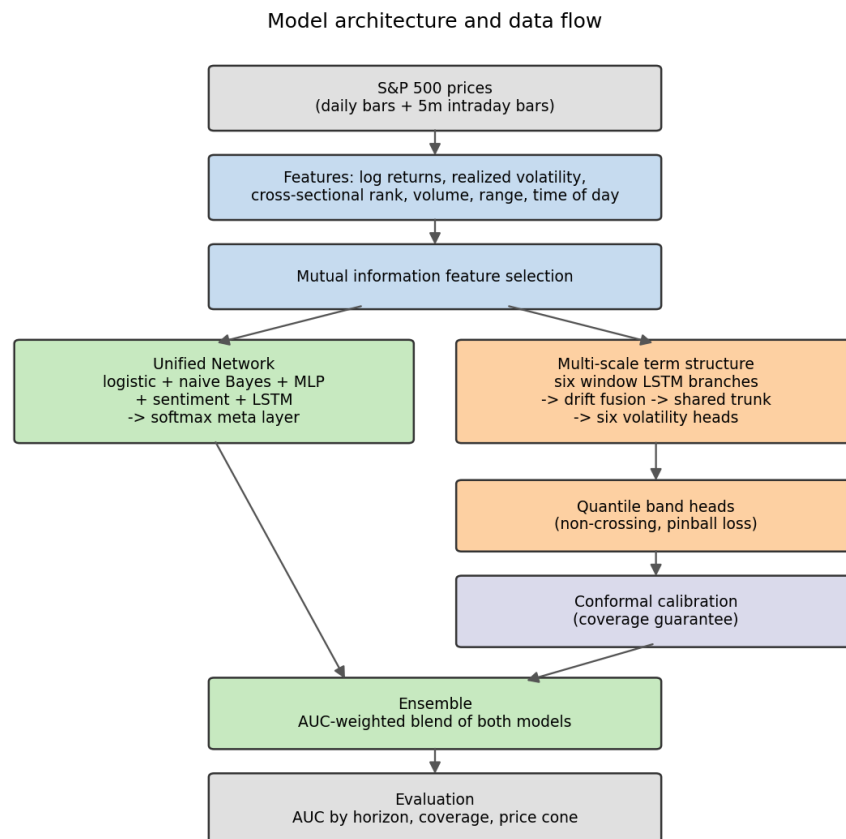


Figure 1: The data flow from raw prices to the calibrated forecast. Features feed a mutual information filter, then two complementary models, then conformal calibration and an ensemble, and finally the horizon by horizon evaluation.

## 2.1 Target Choice That Matches the Signal

The first design decision is to predict volatility rather than direction, because the problem analysis showed that the sign of a move is near a coin flip while the size of a move clusters in time. We make the target market neutral by ranking each stock against its peers on the same date and splitting at the median, so the model learns relative volatility and does not simply detect whether the whole market is calm. The exact label is derived in Section 3.

## 2.2 Multi Scale Term Structure With Drift

Because volatility is predicted with respect to time, the design reads six time windows in parallel, one for each horizon of one, five, ten, thirty, ninety, and one hundred eighty days. Each window is a recurrent branch that summarizes what that scale sees, and the branches are linked by their drift, the difference between adjacent scale embeddings, which measures how the short view departs from the long view. The drift is part of the forward pass, so its gradient trains every branch, and a curvature penalty keeps the six horizon predictions on one smooth term structure. This is the core feature that lets one model learn the whole curve rather than a single horizon, and its formulas are in Section 11.

## 2.3 Complementary Models Under One Roof

No single learner captures every pattern, so the design combines several. The unified network runs a logistic branch, a Gaussian naive Bayes branch, a rectified network, a sentiment branch, and a recurrent branch, then a softmax meta layer learns how much to trust each. The linear branches capture simple monotone effects, the rectified network captures interactions, and the recurrent branch captures the recent path. Each branch is derived in Sections 4 through 9.

## 2.4 A Calibrated Range of Prices, Not a Point

The forecast is only useful if it states how wide the range of outcomes is, so the design adds quantile heads that predict a band of future returns at each horizon. The band is built to be monotone so its edges never cross, it is trained with the pinball loss that targets a chosen quantile, and it is wrapped in conformal calibration so the band covers its nominal rate out of sample with a finite sample guarantee. Turning the return band into prices produces the cone of possible gains and losses. These pieces are derived in Sections 12 and 13.

## 2.5 Robust Training and Honest Evaluation

The design trains every weight with Adam and standardizes features on training rows only, so the test period never leaks into the scaler. It selects features by mutual information to shrink the input, and it judges every horizon by the area under the ROC curve, which measures ranking quality and ignores the threshold. Finally an ensemble blends the two models by the weight that maximizes validation area under the curve. The optimizer, the metric, and the ensemble are derived in Sections 10, 15, and 16.

With the design and its key features in view, the remaining sections build each formula from first principles and work real numbers through it.



### 3 The Model Lifecycle from Data to Tuning

Before the formulas we give the high level view of how the model actually runs end to end, because the mathematics only makes sense inside this loop. The full lifecycle is shown in Figure 2, and it moves from raw data through a training loop with feedback, then to testing, reporting, and tuning, and it cycles back when we modify the design.

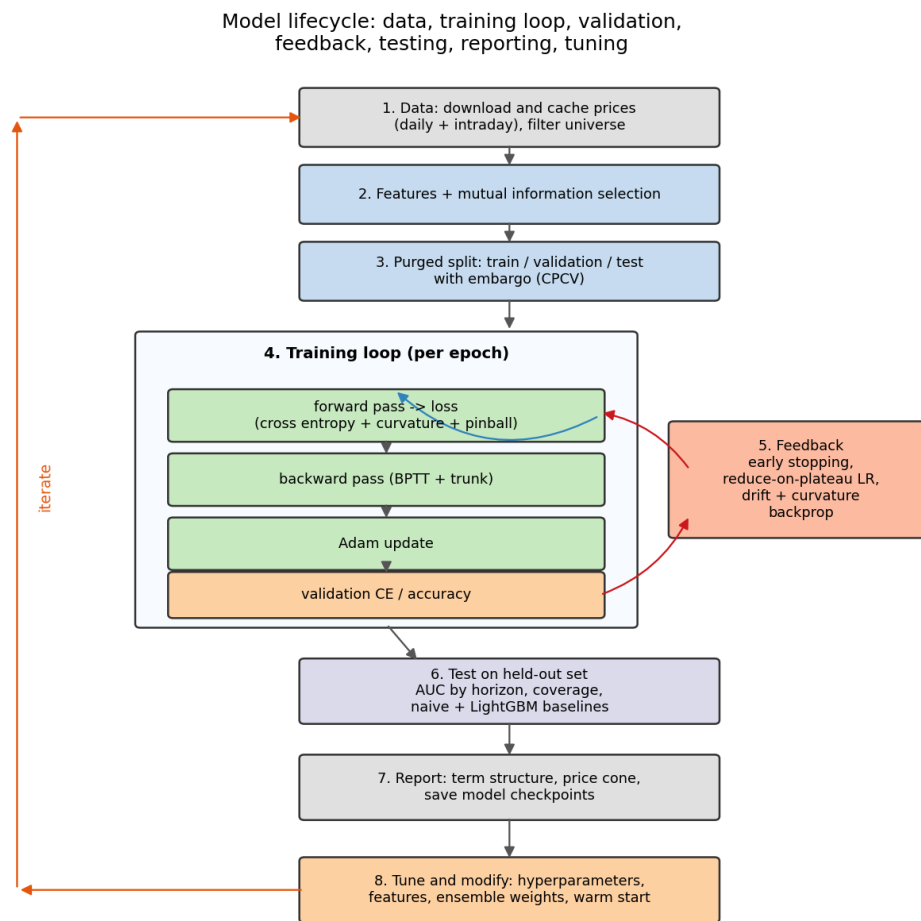


Figure 2: The end to end lifecycle. Data and features feed a purged split, the training loop runs forward, backward, and update steps with validation feedback, and the held out test drives reporting and the tuning that iterates the whole process.

### 3.1 Data

The pipeline downloads daily prices for the S&P 500 universe and separately downloads intraday bars, then caches both so a rerun only fetches what is missing. It filters the universe to liquid names by a minimum median dollar volume, which removes thin stocks whose volatility is dominated by noise rather than signal. From prices it builds returns, realized volatility, volume and range features, and the cross sectional rank that forms the label.

### 3.2 Splitting With a Purge and an Embargo

Financial data is ordered in time and overlapping labels leak information across a naive split, so we split with care. The training block ends before the test block, and we purge the training rows whose forward label window reaches into the test period, then embargo a further gap so no leakage survives. The stronger version is combinatorial purged cross validation, which forms several train and test arrangements while preserving the purge, and it gives the honest estimate that the single split can flatter.

### 3.3 The Training Loop

Inside each epoch the loop shuffles the training rows, forms mini batches, and for each batch runs a forward pass to the loss, a backward pass that backpropagates through the trunk and through time in the recurrent branches, and an Adam update of every weight. The loss is a sum of the cross entropy on the volatility heads, the curvature penalty that smooths the term structure, and the pinball loss on the quantile bands, so one backward pass trains classification and the price bands together.

### 3.4 Validation and Feedback

At the end of every epoch the loop evaluates the loss on a held out validation slice that the gradient never touches, and that number drives two feedback mechanisms. When the validation loss stops improving for a patience window the learning rate is halved, and when it fails to improve for a longer window the loop stops early and restores the best weights. This is the feedback that keeps the model from overfitting the training rows, and its effect is visible as the stepwise drops in the learning rate during a run.

### 3.5 Testing

After training we measure the model on the held out test block that took no part in fitting or in early stopping. We report the area under the ROC curve at every horizon, the coverage of the price bands, and we compare against a naive persistence baseline and a gradient boosted baseline, so a strong number has to beat both a trivial rule and a strong competitor before we trust it.

### 3.6 Reporting and Saving

A run prints the volatility term structure, the area under the curve by horizon, the calibrated band coverage, and an example price cone, all as terminal charts so the shape of the result is visible at a glance. It then saves the trained weights and the feature scaler to a checkpoint, so the model can be reloaded for prediction or combined in the ensemble without retraining.

### 3.7 Tuning and Modifying

The last stage closes the loop. We tune hyperparameters such as the number of epochs, the learning rate schedule, the smoothing weight, the number of tickers, and the intraday bar interval, and we modify the design by adding features, switching the target horizon, refitting the ensemble weight, or warm starting a new run from a saved checkpoint. Each change sends us back to the data and feature stages, which is the outer iterate arrow in the figure.

The sections that follow open each of these boxes and derive the mathematics they contain.

## 4 Data, Returns, and the Prediction Target

The project predicts the volatility of stock returns rather than their direction, because direction is close to unpredictable while volatility clusters in time and carries real signal. Every equation in this document traces back to a function in the code, so the mathematics and the implementation stay in agreement.

The starting point is the daily log return, which turns a ratio of prices into an additive quantity that behaves well under aggregation. For a closing price  $P_t$  We define

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right).$$

The reason for the logarithm is that log returns add across time, so the return over  $h$  days is the sum of the daily log returns, which makes the horizon arithmetic clean.

The forward realized volatility over a horizon of  $h$  trading days is the sample standard deviation of the next  $h$  daily returns. This is the quantity the six horizon heads learn to rank. Writing  $\bar{r}$  for the mean of the window we define

$$\sigma_{t,h} = \sqrt{\frac{1}{h-1} \sum_{k=1}^h (r_{t+k} - \bar{r})^2}, \quad \bar{r} = \frac{1}{h} \sum_{k=1}^h r_{t+k}.$$

The forward shift matters because the label must describe the future, so the window looks forward from the prediction date and never touches past information that the features already contain.

At the one minute and one bar horizon the standard deviation of a single observation is undefined, so we fall back to the absolute forward return, which is the natural one sample scale estimate,

$$\sigma_{t,1} = |r_{t+1}|.$$

### 4.1 Cross Sectional Median Labels

A raw volatility level is not comparable across regimes, because the whole market breathes together. To remove that common component we rank each stock against its peers on the same date and split at the median. For date  $d$  and stock  $i$  the percentile rank is

$$u_{i,d} = \frac{1}{N_d} |\{j : \sigma_{j,d} \leq \sigma_{i,d}\}|,$$

and the binary label is

$$y_{i,d} = \mathbf{1}\left[u_{i,d} \geq \frac{1}{2}\right].$$

Ranking within each date is what makes the target market neutral, so the model learns which names will be more volatile than their peers rather than simply detecting that the whole market is calm or stressed.

The measured volatility term structure of the data is shown in Figure 3. The mean forward realized volatility rises with the horizon, which is the pattern that makes the longer horizons more predictable.

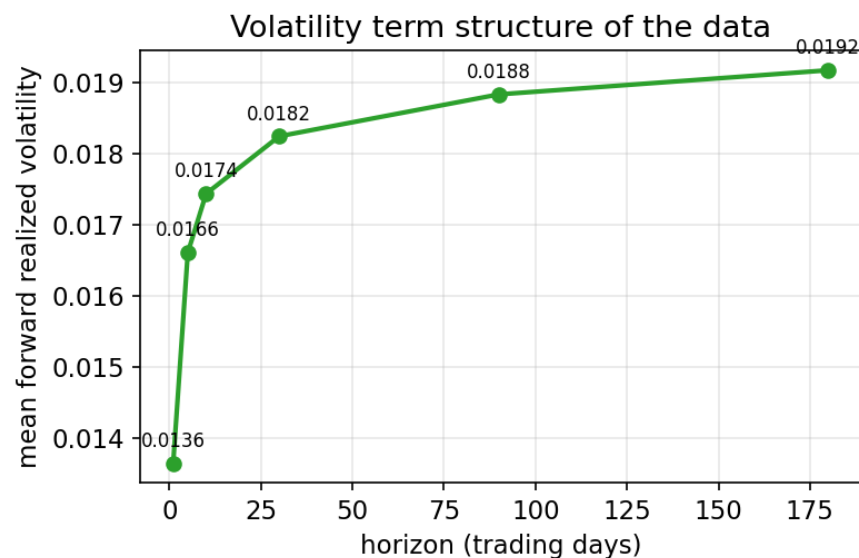


Figure 3: Mean forward realized volatility at each horizon, measured on the daily data. The curve rises from about 0.0136 at one day to about 0.0192 at one hundred eighty days.

## 5 Feature Standardization

Gradient based training is sensitive to the scale of the inputs, because a feature with a large variance dominates the first steps and slows convergence. We therefore standardize every feature to zero mean and unit variance using statistics computed on the training rows only, which prevents the test period from leaking into the scaler. For feature column  $x$  with training mean  $\mu$  and training standard deviation  $s$  the transform is

$$\tilde{x} = \frac{x - \mu}{s + \varepsilon}, \quad \mu = \frac{1}{n} \sum_{i=1}^n x_i, \quad s = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2}.$$

The small constant  $\varepsilon$  guards against a zero variance column, so the division stays finite even when a feature is briefly constant. In the multi scale model this scaler is stored per branch and travels inside the saved checkpoint, so a reloaded model reproduces its training time normalization exactly.

## 6 Information Theory Feature Selection

Before training we keep only the features that share the most information with the target, which shrinks the input and reduces overfitting. The selection criterion is mutual information, so this section derives it from entropy.

### 6.1 Entropy

Entropy measures the average surprise of a distribution, and it is the foundation of the mutual information score. For a discrete variable with probabilities  $p_k$  the entropy in bits is

$$H(X) = - \sum_k p_k \log_2 p_k.$$

The continuous analog replaces the sum with an integral over the density  $f$ , which is the differential entropy

$$h(X) = - \int_{-\infty}^{\infty} f(x) \log_2 f(x) dx.$$

The code cannot integrate an unknown density, so it discretizes the range into bins, estimates each bin probability by its normalized count, and applies the discrete formula. Writing  $c_k$  for the count in bin  $k$  the estimator is

$$\hat{p}_k = \frac{c_k}{\sum_j c_j}, \quad \hat{H}(X) = - \sum_{k:\hat{p}_k > 0} \hat{p}_k \log_2 \hat{p}_k.$$

### 6.2 Mutual Information

Mutual information counts the bits that one variable reveals about another, so it is the right score for feature relevance. It is the Kullback Leibler divergence between the joint distribution and the product of the marginals, which measures how far the two variables are from independence. We start from that definition and reduce it to a difference of entropies.

The reason for starting at the divergence is that independence makes the joint equal to the product of marginals, so the divergence is exactly the information gained by knowing

the joint.

$$\begin{aligned}
I(X, Y) &= \sum_{x,y} p(x, y) \log_2 \frac{p(x, y)}{p(x) p(y)} \\
&= \sum_{x,y} p(x, y) \log_2 p(x, y) - \sum_{x,y} p(x, y) \log_2 p(x) - \sum_{x,y} p(x, y) \log_2 p(y) \\
&= -H(X, Y) + H(X) + H(Y).
\end{aligned}$$

Collecting the terms gives the identity the selector uses, and it is symmetric and nonnegative,

$$\boxed{I(X, Y) = H(X) + H(Y) - H(X, Y).}$$

The implementation estimates  $H(X)$  and  $H(Y)$  from one dimensional histograms and  $H(X, Y)$  from a two dimensional histogram, then keeps the top  $k$  features ranked by  $I(X_j, Y)$ . The clamp at zero absorbs the small negative values that finite bin counts can produce.

We verify the identity on a small joint distribution where the two variables are correlated. Taking the joint mass  $[0.40, 0.10; 0.10, 0.40]$  over two states each the marginals are uniform, so both marginal entropies are one bit. Working the numbers through the identity gives

$$\begin{aligned}
H(X) &= 1.0000, & H(Y) &= 1.0000, & H(X, Y) &= 1.7219, \\
I(X, Y) &= 1.0000 + 1.0000 - 1.7219 = 0.2781 \text{ bits.}
\end{aligned}$$

The positive value confirms the two variables share information, and the ranking this produces on the real features is shown in Figure 4, where a long horizon volatility feature carries the most information about the target.



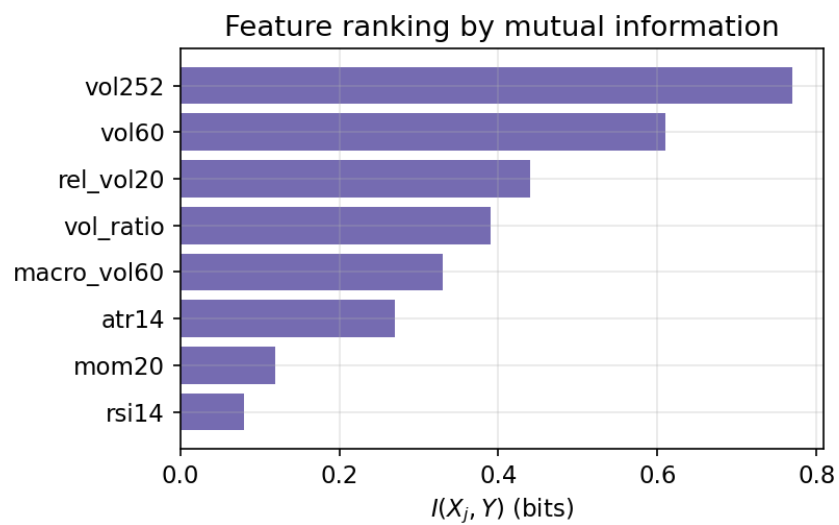


Figure 4: Features ranked by mutual information with the volatility label. The realized volatility over a long window carries the most bits, near 0.77.

## 7 Branch A: Logistic Regression

The first branch is a logistic model, and it is the cleanest place to derive the gradient that reappears in every later layer. The model passes a linear score through the logistic function, which maps the real line to a probability,

$$p = \sigma(z), \quad z = w^\top x + b, \quad \sigma(z) = \frac{1}{1 + e^{-z}}.$$

We will need the derivative of the logistic function, and it has a compact form that falls out of the quotient rule. Differentiating and then regrouping so the result is expressed in the output itself gives

$$\begin{aligned} \sigma'(z) &= \frac{d}{dz} (1 + e^{-z})^{-1} = \frac{e^{-z}}{(1 + e^{-z})^2} \\ &= \frac{1}{1 + e^{-z}} \cdot \frac{e^{-z}}{1 + e^{-z}} = \sigma(z) (1 - \sigma(z)). \end{aligned}$$

The label is Bernoulli given the features, so the likelihood of one observation is  $p^y(1 - p)^{1-y}$ . Training maximizes the log likelihood, which is the same as minimizing its negative, and that negative is the binary cross entropy used in the code,

$$\mathcal{L} = -\left[ y \ln p + (1 - y) \ln(1 - p) \right].$$

The key result is the derivative of this loss with respect to the score  $z$ , because it collapses to a simple residual. We apply the chain rule through  $p$  and then substitute the logistic derivative so the two fractions cancel.

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial z} &= \frac{\partial \mathcal{L}}{\partial p} \frac{\partial p}{\partial z} = \left( -\frac{y}{p} + \frac{1-y}{1-p} \right) \sigma(z)(1 - \sigma(z)) \\ &= \left( \frac{p-y}{p(1-p)} \right) p(1-p) = p - y. \end{aligned}$$

The gradient with respect to the weights follows immediately from the chain rule, and it is the residual times the input,

$$\boxed{\frac{\partial \mathcal{L}}{\partial w} = (p - y) x, \quad \frac{\partial \mathcal{L}}{\partial b} = p - y.}$$

We work three scores through the branch to make the mapping concrete. The values below come straight from evaluating the logistic function and its loss, and they are plotted

in Figure 5.

$$\begin{aligned} z = -2 &\Rightarrow p = 0.1192, & \sigma'(z) &= 0.1050, & \mathcal{L}(y=1) &= 2.1269, \\ z = 0 &\Rightarrow p = 0.5000, & \sigma'(z) &= 0.2500, & \mathcal{L}(y=1) &= 0.6931, \\ z = 2 &\Rightarrow p = 0.8808, & \sigma'(z) &= 0.1050, & \mathcal{L}(y=1) &= 0.1269. \end{aligned}$$

The loss falls as the score moves toward the correct side, and the slope is largest at  $z = 0$  where the model is least certain, which is where learning is fastest.

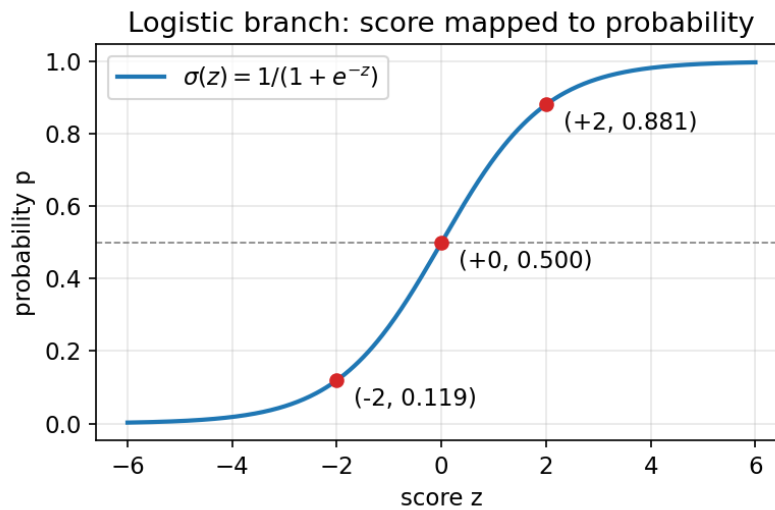


Figure 5: The logistic map with the three worked scores marked. The dashed line is the decision boundary at probability one half.

## 8 Branch B: Gaussian Naive Bayes

The second branch assumes each feature is Gaussian within a class and that the features are conditionally independent, which is why it is called naive. This section shows why that assumption produces a linear score, which is the form the code implements after standardizing by learned class statistics.

The class conditional density for a single feature is Gaussian, and its integral over the whole line is one, which is the normalization that makes it a proper density. We record that normalization because it justifies dropping the constant term later,

$$\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi} s} \exp\left(-\frac{(x - \mu)^2}{2s^2}\right) dx = 1.$$

By conditional independence the joint class conditional density is the product over features, so the log likelihood is a sum of quadratics. Writing the log posterior odds between class one and class zero, and using a shared variance so the quadratic terms cancel, the score becomes linear in  $x$ .

$$\begin{aligned} \ln \frac{P(y = 1 \mid x)}{P(y = 0 \mid x)} &= \ln \frac{\pi_1}{\pi_0} + \sum_j \left[ -\frac{(x_j - \mu_{1j})^2}{2s_j^2} + \frac{(x_j - \mu_{0j})^2}{2s_j^2} \right] \\ &= \ln \frac{\pi_1}{\pi_0} + \sum_j \frac{\mu_{1j} - \mu_{0j}}{s_j^2} x_j + \text{const.} \end{aligned}$$

Because the log odds is linear, the class probability is a logistic function of a linear score, which is exactly the branch the code builds after it whitens the input with a learned mean and a learned log variance. Writing the whitened input as  $\tilde{x} = (x - \mu)/s$  the branch output is

$$\boxed{a_{\text{nb}} = \sigma(W_{\text{nb}}^\top \tilde{x} + b_{\text{nb}}), \quad s = e^\ell + \varepsilon.}$$

The parametrization through  $\ell = \ln s$  keeps the standard deviation strictly positive under unconstrained gradient steps, so the optimizer never drives a variance to a negative value.

## 9 Branch C: The Multilayer Perceptron

The third branch is a fully connected network with rectified linear activations, and it supplies the nonlinear capacity the linear branches lack. Each layer applies an affine map and then a rectifier, which zeroes the negative part,

$$Z^{(l)} = A^{(l-1)}W^{(l)} + b^{(l)}, \quad A^{(l)} = \max(0, Z^{(l)}).$$

The rectifier has a simple derivative that acts as a gate during backpropagation, because it passes the upstream gradient only where the preactivation was positive,

$$\frac{\partial A^{(l)}}{\partial Z^{(l)}} = \mathbf{1}[Z^{(l)} > 0].$$

Backpropagation through one layer therefore multiplies the incoming gradient by this gate and then pushes it through the linear map. Writing  $\delta$  for the gradient with respect to the activation the three local gradients are

$$\begin{aligned} \delta_Z &= \delta_A \odot \mathbf{1}[Z^{(l)} > 0], \\ \frac{\partial \mathcal{L}}{\partial W^{(l)}} &= (A^{(l-1)})^\top \delta_Z, \\ \delta_A^{(l-1)} &= \delta_Z (W^{(l)})^\top. \end{aligned}$$

Dropout multiplies the activations by a scaled Bernoulli mask during training, which discourages the network from relying on any single unit. The mask divides by the keep probability so the expected activation is unchanged, and it is replayed in the backward pass so the gradient matches the forward computation.

## 10 Branch E: The LSTM and Backpropagation Through Time

The recurrent branch reads a lookback window of feature vectors and summarizes it into one embedding, and it is the workhorse of the multi scale model. The long short term memory cell keeps a protected cell state that carries information across many steps, which is what lets it see long horizons without the gradient vanishing.

### 10.1 Forward Recurrence

At each step the cell computes four gates from the current input and the previous hidden state, then updates the cell and the hidden state. The forget gate decides what to keep, the input gate and the candidate decide what to add, and the output gate decides what to expose. With  $\sigma$  the logistic function the recurrence is

$$\begin{aligned} [z_f; z_i; z_g; z_o] &= W^\top x_t + U^\top h_{t-1} + b, \\ f_t &= \sigma(z_f), \quad i_t = \sigma(z_i), \quad g_t = \tanh(z_g), \quad o_t = \sigma(z_o), \\ c_t &= f_t \odot c_{t-1} + i_t \odot g_t, \quad h_t = o_t \odot \tanh(c_t). \end{aligned}$$

The additive cell update is the crucial design choice, because the path from  $c_{t-1}$  to  $c_t$  has a derivative equal to the forget gate rather than a product of small activation slopes, so gradients survive over long windows.

### 10.2 Backward Recurrence

Backpropagation through time walks the sequence in reverse and accumulates the gradient of the loss into the weights. The derivation below matches the loop in the code exactly. We begin at step  $t$  with the incoming hidden gradient  $\partial\mathcal{L}/\partial h_t$  and the carried cell gradient from the future, and we use the tanh and logistic derivatives at each gate.

The output gate feeds the hidden state through a tanh, so its gradient comes from differentiating  $h_t = o_t \odot \tanh(c_t)$ ,

$$\begin{aligned} \delta o_t &= \delta h_t \odot \tanh(c_t), \\ \delta c_t &= \delta h_t \odot o_t \odot (1 - \tanh^2(c_t)) + \delta c_t^{\text{future}}. \end{aligned}$$

The cell update distributes its gradient to the forget gate, the input gate, the candidate,

and the previous cell, because each is a factor in an elementwise product,

$$\begin{aligned}\delta f_t &= \delta c_t \odot c_{t-1}, & \delta i_t &= \delta c_t \odot g_t, \\ \delta g_t &= \delta c_t \odot i_t, & \delta c_{t-1} &= \delta c_t \odot f_t.\end{aligned}$$

Each gate gradient then passes through its own activation derivative, which is the logistic slope for the three sigmoid gates and the tanh slope for the candidate,

$$\begin{aligned}\delta z_f &= \delta f_t \odot f_t \odot (1 - f_t), & \delta z_i &= \delta i_t \odot i_t \odot (1 - i_t), \\ \delta z_g &= \delta g_t \odot (1 - g_t^2), & \delta z_o &= \delta o_t \odot o_t \odot (1 - o_t).\end{aligned}$$

Stacking the four preactivation gradients into one vector  $\delta z_t$  gives the weight gradients as outer products and the recurrent gradient that flows to the previous step,

$$\boxed{\frac{\partial \mathcal{L}}{\partial W} = \sum_t x_t \delta z_t^\top, \quad \frac{\partial \mathcal{L}}{\partial U} = \sum_t h_{t-1} \delta z_t^\top, \quad \frac{\partial \mathcal{L}}{\partial b} = \sum_t \delta z_t, \quad \delta h_{t-1} = U \delta z_t.}$$

## 11 The Softmax Meta Layer and Cross Entropy

The branch outputs are concatenated and combined by a softmax classifier that learns how much to trust each branch. The softmax turns a score vector into a distribution over classes, and it is invariant to a constant shift, which the code exploits for numerical stability by subtracting the row maximum. For class  $k$  the probability is

$$\hat{y}_k = \frac{e^{z_k}}{\sum_j e^{z_j}}.$$

The training loss is categorical cross entropy against the one hot label, which is the negative log likelihood of the correct class,

$$\mathcal{L} = - \sum_k y_k \ln \hat{y}_k.$$

The reason the softmax and cross entropy are paired is that their composition has the same clean residual gradient as the logistic case, which we now derive. The derivative of the softmax has two cases depending on whether the index matches, and combining them collapses the sum.

$$\begin{aligned} \frac{\partial \hat{y}_k}{\partial z_m} &= \hat{y}_k (\mathbf{1}[k = m] - \hat{y}_m), \\ \frac{\partial \mathcal{L}}{\partial z_m} &= - \sum_k \frac{y_k}{\hat{y}_k} \hat{y}_k (\mathbf{1}[k = m] - \hat{y}_m) = -y_m + \hat{y}_m \sum_k y_k. \end{aligned}$$

Because the label is one hot its entries sum to one, so the final gradient is the difference between the predicted and the true distribution,

$$\boxed{\frac{\partial \mathcal{L}}{\partial z_m} = \hat{y}_m - y_m.}$$

We run one score vector through the layer to show the residual gradient in numbers. Taking the scores  $z = [2.0, 0.5, -1.0]$  and the true class zero the softmax, the loss, and the gradient are

$$\begin{aligned} \hat{y} &= [0.7856, 0.1753, 0.0391], & \sum_k \hat{y}_k &= 1.0000, \\ \mathcal{L} &= -\ln(0.7856) = 0.2413, \\ \hat{y} - y &= [-0.2144, 0.1753, 0.0391]. \end{aligned}$$



The gradient pushes the score of the true class up and the other two down, which is the direction that lowers the loss.

## 12 The Adam Optimizer

All weights are trained with Adam, which adapts a separate step size to each parameter using running estimates of the first and second moments of the gradient. This section derives the bias correction, which is the part that is easy to get wrong.

Adam keeps exponential moving averages of the gradient and its square, so recent gradients count more than old ones. With decay rates  $\beta_1$  and  $\beta_2$  the updates are

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t, \quad v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2.$$

Because the averages start at zero they are biased toward zero early in training, and we quantify that bias by unrolling the recursion into a geometric sum. Taking expectations and assuming the second moment is roughly stationary over the window gives

$$v_t = (1 - \beta_2) \sum_{k=1}^t \beta_2^{t-k} g_k^2,$$

$$\mathbb{E}[v_t] \approx \mathbb{E}[g^2] (1 - \beta_2) \sum_{k=1}^t \beta_2^{t-k} = \mathbb{E}[g^2] (1 - \beta_2^t).$$

The factor  $1 - \beta_2^t$  is exactly the shrinkage we must undo, so dividing by it removes the bias, and the same argument applies to the first moment. The bias corrected estimates and the parameter update are

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}, \quad \hat{v}_t = \frac{v_t}{1 - \beta_2^t}, \quad \boxed{\theta_t = \theta_{t-1} - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \varepsilon}.$$

The code also adds an  $L_2$  term to the gradient of the weight matrices before the moment updates, which pulls the weights toward zero and controls their magnitude.

We trace three steps on a constant gradient  $g = 0.5$  to show the bias correction at work. Early on the raw moments  $m$  and  $v$  are far below their long run values, yet the corrected estimates  $\hat{m}$  and  $\hat{v}$  already sit at the true levels, so the step size is stable from the first iteration.

| $t$ | $m_t$  | $v_t$    | $\hat{m}_t$ | $\hat{v}_t$ | $\theta_t$ |
|-----|--------|----------|-------------|-------------|------------|
| 1   | 0.0500 | 0.000250 | 0.5000      | 0.250000    | 0.999000   |
| 2   | 0.0950 | 0.000500 | 0.5000      | 0.250000    | 0.998000   |
| 3   | 0.1355 | 0.000749 | 0.5000      | 0.250000    | 0.997000   |

The corrected second moment stays at 0.25, which is the square of the gradient, so the update reduces to a clean step of size  $\eta$  toward the minimum.

## 13 Multi Scale Fusion and the Drift Gradient

The multi scale model runs six LSTM branches over windows of one, five, ten, thirty, ninety, and one hundred eighty days, and it links them by their drift. The drift between adjacent scales is the difference of their embeddings, and it measures how the short view departs from the long view. Writing  $e_b$  for the embedding of branch  $b$  the drift and the fused vector are

$$\Delta_k = e_{k+1} - e_k, \quad \text{fuse} = [e_1, \dots, e_B, \Delta_1, \dots, \Delta_{B-1}].$$

Because the drift is a real part of the forward pass, its gradient must flow back into the branches that formed it. Differentiating  $\Delta_k = e_{k+1} - e_k$  shows that a unit of drift gradient adds to the later embedding and subtracts from the earlier one, which is the sign pattern in the backward split,

$$\boxed{\frac{\partial \mathcal{L}}{\partial e_{k+1}} + = \frac{\partial \mathcal{L}}{\partial \Delta_k}, \quad \frac{\partial \mathcal{L}}{\partial e_k} - = \frac{\partial \mathcal{L}}{\partial \Delta_k}.$$

This is the precise sense in which model drift across time windows is backpropagated as context, because the disagreement between scales feeds gradient into every branch that produced it.

### 13.1 The Curvature Coupling Penalty

The six horizon probabilities should trace a smooth term structure rather than jump from one horizon to the next, so we penalize the roughness of the curve. The natural smoothness measure in continuous form is the integral of the squared second derivative, which is large where the curve bends sharply,

$$R_{\text{cont}} = \int (f''(x))^2 dx.$$

The horizons are discrete, so we replace the second derivative with the second difference, which is the discrete Laplacian of the probability curve. Averaging over the interior horizons and the batch of  $N$  rows the penalty is

$$R = \frac{1}{N(B-2)} \sum_n \sum_{b=2}^{B-1} (P_{n,b+1} - 2P_{n,b} + P_{n,b-1})^2.$$

The gradient of this penalty spreads each second difference back onto the three horizons that formed it with weights one, minus two, and one, which is the stencil applied in the code.

Writing  $D_{n,b} = P_{n,b+1} - 2P_{n,b} + P_{n,b-1}$  the contribution to horizon  $b$  is

$$\frac{\partial R}{\partial P_{n,b}} = \frac{2}{N(B-2)}(D_{n,b-1} - 2D_{n,b} + D_{n,b+1}).$$

This term is added to the classification gradient after being routed through the logistic slope  $P(1-P)$ , so the smoothing acts on the scores that produce the probabilities.

The curvature term measures how sharply the area under the curve bends across horizons. Working the second difference through the measured daily term structure the interior bends are small, which confirms the curve is already nearly smooth,

at 5d :  $-0.0725$ , at 10d :  $+0.0074$ , at 30d :  $-0.0201$ , at 90d :  $-0.0246$ .

Figure 6 shows the area under the curve at every horizon for both the daily and the intraday scales, and both rise with the horizon exactly as the volatility clustering argument predicts.

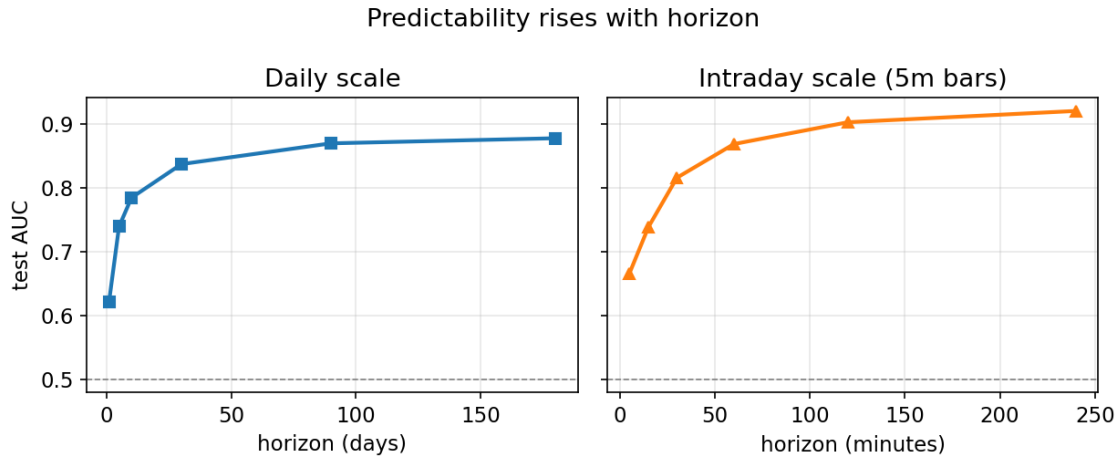


Figure 6: Test area under the curve by horizon. The daily scale climbs from about 0.62 at one day to about 0.88 at the long horizons, and the intraday scale climbs from about 0.67 at five minutes to about 0.92 at four hours.

## 14 Quantile Price Bands

The model also predicts a band of possible future returns at each horizon, not just a volatility label, so it can express a price range. The bands are trained with quantile regression, and this section derives why the check loss recovers a quantile and how the monotone construction keeps the band edges from crossing.

### 14.1 The Pinball Loss and the Quantile

For a target level  $\tau$  the pinball loss weights positive and negative errors differently, which tilts the optimum toward the  $\tau$  quantile. With residual  $u = y - q$  the loss is

$$\rho_\tau(u) = \max(\tau u, (\tau - 1)u) = u(\tau - \mathbf{1}[u < 0]).$$

We now show that the expected pinball loss is minimized at the quantile, which is the reason this loss is used. We split the expectation at  $q$  so the absolute value opens into two integrals, then differentiate with respect to  $q$  using the Leibniz rule. The prediction  $q$  appears only in the limits and in the integrands, and the boundary terms vanish because the integrand is zero at  $u = 0$ .

$$\begin{aligned} \mathbb{E}[\rho_\tau(Y - q)] &= \tau \int_q^\infty (y - q) f(y) dy + (1 - \tau) \int_{-\infty}^q (q - y) f(y) dy, \\ \frac{d}{dq} \mathbb{E}[\rho_\tau(Y - q)] &= -\tau \int_q^\infty f(y) dy + (1 - \tau) \int_{-\infty}^q f(y) dy \\ &= -\tau (1 - F(q)) + (1 - \tau) F(q) = F(q) - \tau. \end{aligned}$$

Setting the derivative to zero forces the cumulative probability at the optimum to equal  $\tau$ , so the minimizer is exactly the  $\tau$  quantile,

$$\boxed{F(q^*) = \tau \implies q^* = F^{-1}(\tau).}$$

The subgradient used in training follows from the same expression, and it is a constant on each side of the residual. With  $e = y - q$  the derivative with respect to the prediction  $q$  is

$$\frac{\partial \rho_\tau}{\partial q} = \begin{cases} -\tau, & e > 0, \\ 1 - \tau, & e \leq 0, \end{cases}$$

which is the exact branch the code writes when it forms the quantile gradient.

We confirm the theory by minimizing the empirical pinball loss over a sample of twenty thousand standard normal draws, and the minimizer lands on the sample quantile at every level, which is the guarantee the derivation promised.

$$\begin{aligned}\tau = 0.05 &\Rightarrow q^* = -1.620, & \text{sample quantile} &= -1.621, \\ \tau = 0.50 &\Rightarrow q^* = -0.010, & \text{sample quantile} &= -0.006, \\ \tau = 0.95 &\Rightarrow q^* = +1.630, & \text{sample quantile} &= +1.633.\end{aligned}$$

The shape of the loss for three levels is shown in Figure 7, where the tilt of each arm sets which quantile the optimum finds.

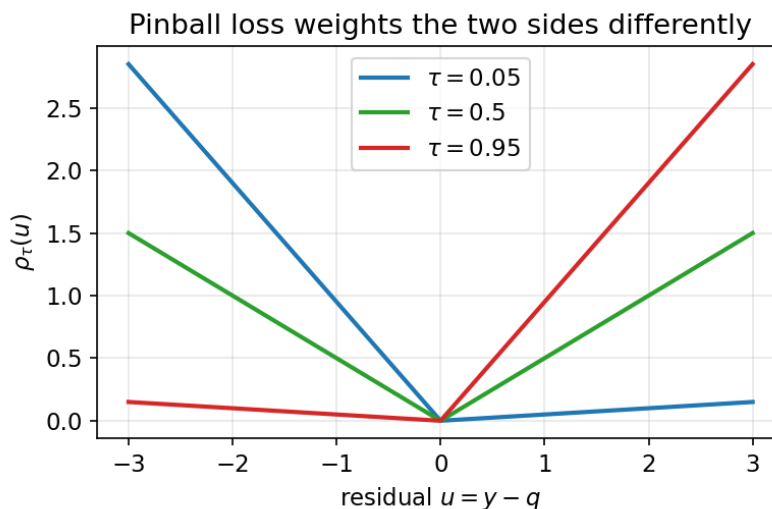


Figure 7: The pinball loss for three target levels. The steeper arm penalizes the side that must be rare, which pushes the optimum to the requested quantile.

## 14.2 The Monotone Non Crossing Construction

A set of quantiles is only meaningful if the lower ones stay below the higher ones, so we build the band to be monotone by construction rather than hoping the optimizer keeps it ordered. The lowest quantile is a free level, and each higher quantile adds a strictly positive increment produced by a softplus, whose output is always positive,

$$\text{softplus}(z) = \ln(1 + e^z) > 0.$$

Writing the raw head outputs as  $r_0, r_1, \dots, r_{Q-1}$  the quantiles are a running sum of these

increments, which is nondecreasing because every term added is positive,

$$q_0 = r_0, \quad q_j = r_0 + \sum_{m=1}^j \text{softplus}(r_m), \quad j = 1, \dots, Q-1.$$

The gradient must pass through this cumulative sum, so a change in one increment affects every quantile at or above its index. That makes the gradient with respect to increment  $m$  a reverse cumulative sum of the upstream quantile gradients, scaled by the softplus derivative, which is the logistic function,

$$\boxed{\frac{\partial \mathcal{L}}{\partial r_m} = \sigma(r_m) \sum_{j \geq m} \frac{\partial \mathcal{L}}{\partial q_j}.$$

This is exactly the reverse cumulative sum times  $\sigma(\text{raw})$  that the backward pass computes before routing the gradient into the shared trunk, so the price bands and the volatility heads train the same representation.



## 15 Conformal Calibration of the Bands

Quantile heads are not guaranteed to cover their nominal rate out of sample, so we wrap them in conformalized quantile regression, which widens the band by a data driven amount that restores coverage with a finite sample guarantee. This section states the score, the width, and the coverage proof.

On a held out calibration set we measure how far each true return falls outside its predicted band, which is zero when the point is inside and positive when it is outside. With lower and upper predictions  $q_{lo}$  and  $q_{hi}$  the conformity score is

$$s_i = \max(q_{lo}(x_i) - y_i, y_i - q_{hi}(x_i)).$$

The band widening is the empirical quantile of these scores at a level chosen so the finite sample bound holds. For  $n$  calibration points and target miscoverage  $\alpha$  the index and the widening are

$$k = \lceil (1 - \alpha)(n + 1) \rceil, \quad \delta = s_{(k)},$$

where  $s_{(k)}$  is the  $k$  th smallest score, clamped to be nonnegative. The calibrated band is then  $[q_{lo} - \delta, q_{hi} + \delta]$ .

The coverage guarantee rests on exchangeability of the calibration and test points, because then the rank of the test score among the calibration scores is uniform. A new point is covered exactly when its score does not exceed  $\delta$ , so its miscoverage probability is the chance that its score lands above the  $k$  th order statistic,

$$\mathbb{P}(s_{n+1} > s_{(k)}) = \frac{n + 1 - k}{n + 1} \leq \alpha.$$

Taking the complement gives the finite sample coverage bound that the calibration targets, and it holds for any underlying model,

$$\boxed{\mathbb{P}(Y_{n+1} \in [q_{lo} - \delta, q_{hi} + \delta]) \geq 1 - \alpha.}$$

We demonstrate the widening on a deliberately narrow band. Starting from a band that covers only 0.683 of a normal sample, five hundred calibration scores set the widening to  $\delta = 0.669$  at the index  $k = 451$ , and the calibrated band then covers 0.907 of fresh data against the target of 0.90. Figure 8 shows the distribution of the scores with the chosen widening marked.

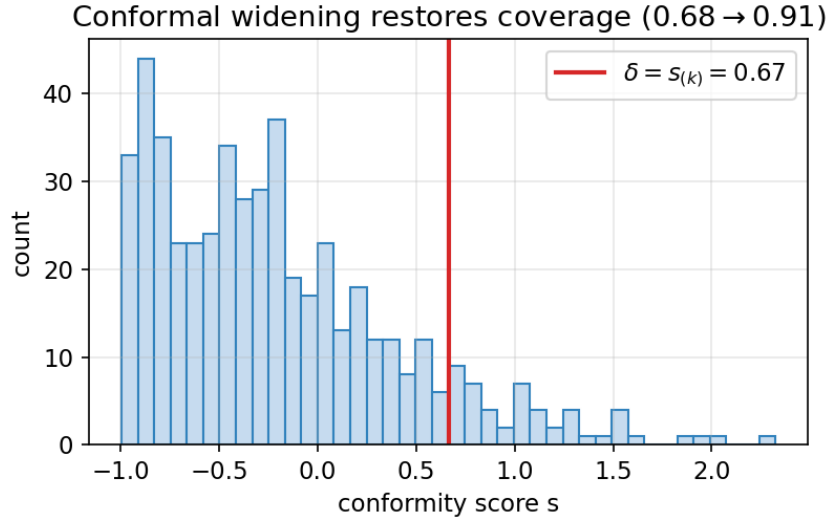


Figure 8: Conformity scores on the calibration set. The vertical line is the order statistic  $\delta = s_{(k)}$  that widens the band, and it lifts coverage from 0.68 to 0.91.

The example price cone reported at the end of a run turns these return quantiles into prices through  $P = P_0 e^q$ , because the model works in log return space where the bands are symmetric and additive.

The grade of the cone, meaning how fast it widens with the horizon, is set by integrating the variance under the volatility curve rather than by any single volatility. Returns over disjoint days are close to uncorrelated, so variances add across time, and the variance of the cumulative return to horizon  $h$  is the area under the instantaneous variance curve. The half width of the band scales with the square root of that integral,

$$\sigma_{\text{cum}}(h) = \sqrt{\int_0^h \sigma^2(\tau) d\tau} \approx \sqrt{\sum_{k=1}^h \sigma_k^2},$$

where  $\sigma_k$  is the measured volatility on day  $k$  taken from the term structure. Because the measured volatility rises with the horizon the integrated variance grows faster than a flat volatility would predict, so the cone is steeper and it matches the grade of the data. A flat one day volatility is the special case where  $\sigma(\tau)$  is constant, which understates the area and flattens the cone.

Figure 9 shows the cone interacting with the data projection. We build the band from the measured volatility term structure using the normal quantiles, and we project the data by simulating four thousand price paths under the same rising volatility schedule. The shaded model cone and the empirical quantiles of the simulated paths track each other closely, and

the fraction of paths that finish inside the five to ninety five band is 0.901, which matches the nominal ninety percent. Working the one hundred eighty day edges through  $P = 100e^q$  gives

|                                |                                |
|--------------------------------|--------------------------------|
| 5th \$66.36 (data \$66.32),    | 25th \$84.52 (data \$84.25),   |
| 50th \$100.00 (data \$100.67), | 75th \$118.32 (data \$119.74), |
| 95th \$150.70 (data \$150.32). |                                |

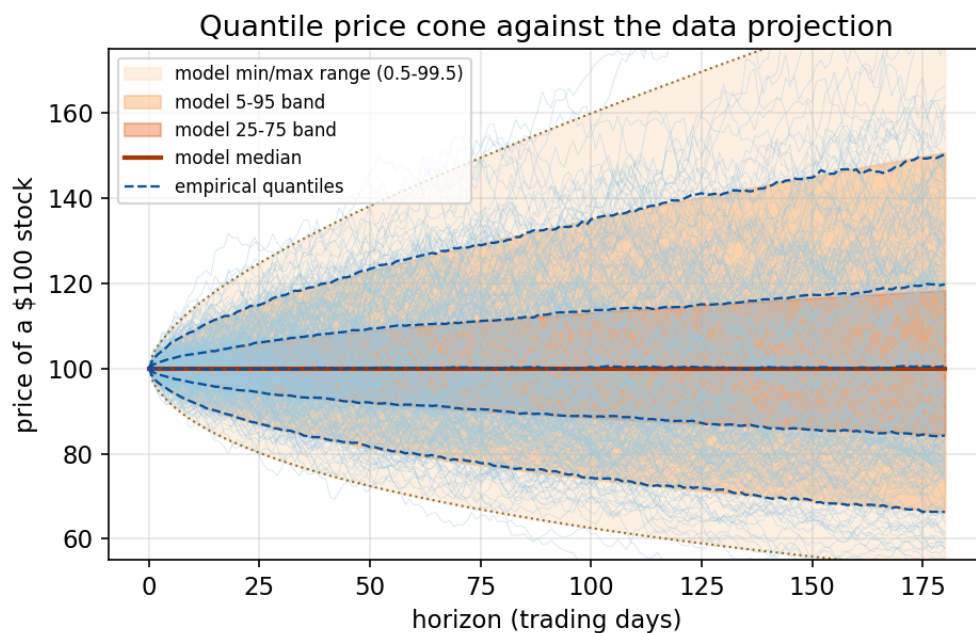


Figure 9: The quantile price cone against the data projection. The light lines are simulated price paths, the shaded regions are the model bands, and the dashed lines are the empirical quantiles of the paths. The cone widens as the square root of the integrated variance under the volatility curve and contains the paths at the nominal rate.

## 16 Evaluation by the Area Under the ROC Curve

We judge every horizon by the area under the receiver operating characteristic curve, because it measures ranking quality and is invariant to the threshold and to class imbalance. The curve plots the true positive rate against the false positive rate as the decision threshold sweeps, and the area under it has a probabilistic meaning that leads to a fast estimator.

The area is the integral of the true positive rate over the false positive rate, and that integral equals the probability that a random positive is scored above a random negative,

$$\text{AUC} = \int_0^1 \text{TPR} d(\text{FPR}) = \mathbb{P}(S^+ > S^-).$$

Estimating that probability by counting ordered pairs would cost a quadratic number of comparisons, so we use the Mann Whitney identity, which reads the same count off the ranks. Sorting all scores and summing the ranks of the positives gives the estimator the code implements,

$$\widehat{\text{AUC}} = \frac{R^+ - \frac{1}{2} n_+ (n_+ + 1)}{n_+ n_-},$$

where  $R^+$  is the sum of the ranks of the positive examples and  $n_+$  and  $n_-$  are the counts of positives and negatives. The subtracted term removes the ranks that positives contribute among themselves, so what remains counts only the positive over negative wins.

We check the identity on a labeled sample of forty scores. The rank based formula and a direct count of ordered pairs give the same value, which confirms the two views agree,

$$\widehat{\text{AUC}}_{\text{ranks}} = 0.6522 = \widehat{\text{AUC}}_{\text{pairs}} = 0.6522.$$

The corresponding curve is shown in Figure 10, where the area above the diagonal is the reported score.

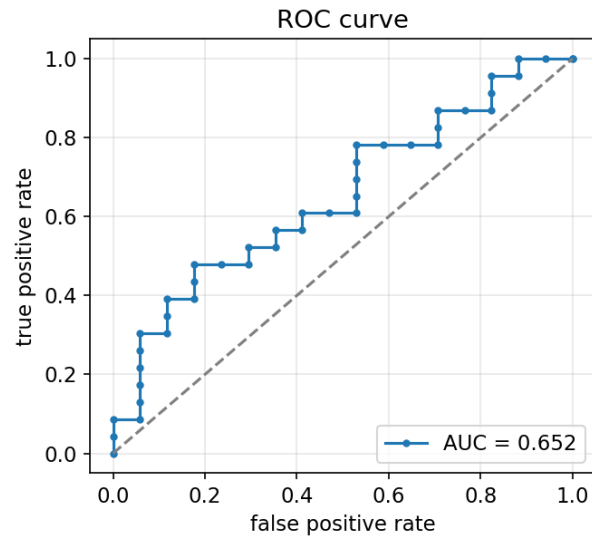


Figure 10: An ROC curve for the worked sample. The diagonal is the chance line, and the area under the curve equals the rank based estimate of 0.652.

## 17 The Ensemble

The final prediction blends the cross sectional volatility model with the multi scale term structure model, because they read different views of the same event and their errors are partly independent. Both output a probability that the next horizon volatility is above the cross sectional median, so a convex combination is a valid probability,

$$p_{\text{ens}} = a p_{\text{ucn}} + (1 - a) p_{\text{ms}}, \quad a \in [0, 1].$$

The blend weight is chosen to maximize validation area under the curve over a grid, which picks the mix that ranks best out of sample,

$$a^* = \arg \max_{a \in [0, 1]} \text{AUC}(y, a p_{\text{ucn}} + (1 - a) p_{\text{ms}}).$$

When only one model has inputs available the search degenerates to that model alone, so the ensemble always reduces gracefully to whichever component is present.

## 18 Summary of the Pipeline

The project chains the pieces above into one estimator. We compute log returns and forward realized volatility, rank the volatility within each date to form market neutral labels, standardize the features on training rows, and keep the features with the most mutual information with the target. The unified network combines a logistic branch, a Gaussian naive Bayes branch, a rectified network, a sentiment branch, and an LSTM branch under a softmax meta layer, and it is trained by Adam with cross entropy. The multi scale model adds six window LSTM branches linked by their drift, six volatility heads coupled by a curvature penalty, and quantile heads that produce conformally calibrated price bands. We score every horizon by the area under the ROC curve and blend the two models into a single calibrated probability. Every equation in this document is the one the code evaluates, so the statistics and the software describe the same model.